SHIP TYPES 1.4 · 10 MIN

Bulk Carriers

The workhorse of deadweight shipping: layout, size classes, strength and rules · Session W1.4 · Not in the Evaluation Test

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Holds Shaped by the Cargo

Section

Single deck; hopper tanks low in the wings and topside tanks under the deck

The hoppers make cargo slide to the hatch; the topsides stop grain shifting and carry ballast

Along the ship

Five to nine holds forward of an aft engine room and accommodation

Side-rolling or folding hatch covers; cranes on the smaller sizes

Form

Full and slow: CB 0.72–0.86, Fn about 0.15–0.18

DWT/Δ 0.74–0.87 (Table 2.1)

The design starts from deadweight (Table 1.8): Δ from DWT, then L, CB, B and T, as for our reference tanker. Container ships start from TEU (Table 1.9).

Size Classes Follow Ports and Canals

Class	Deadweight, about	What sets it
Handysize	15,000–40,000 t	Small ports; usually geared
Handymax, Supramax	40,000–65,000 t	Geared, flexible trades
Panamax, Kamsarmax	65,000–85,000 t	Panama locks (old), Kamsar port in Guinea
Capesize	100,000–200,000 t	Too large for Panama; most go round the Cape when loaded
Very large ore carriers	200,000–400,000 t	Dedicated iron-ore terminals

> Class names are market labels with blurred edges; designers work from the port and canal limits behind them (Ship types 1.2).

Dense Cargo Loads the Hull Unevenly

Ore in alternate holds

Heavy ore fills a hold only partly, so ore ships often load every other hold

Full and empty holds side by side: high shear forces and local loads

Grain

Light and free-flowing: it can shift in a seaway and heel the ship

Topside tanks and the IMO Grain Code limit the heeling moment

Steel weight

Ore strengthening adds steel: Danckwardt's correction is up to about 6% of DWT

Large bulk carriers are among the lowest in WST/LBD, about 76 kg/m³

Losses Led to New Rules

When	What happened	What changed
1980	Derbyshire, a large ore-bulk-oil carrier, lost in a typhoon with 44 people	Investigations pointed to forward hatch covers and openings
Around 1990	A run of bulk carrier losses, many with no survivors	Investigations into the losses
1997	IMO SOLAS conference; IACS	SOLAS Chapter XII and IACS unified requirements for bulk carriers
2006, 2015	Class societies harmonize their rules	Common Structural Rules for bulk carriers and tankers

> The lesson for any designer: the rule book is a record of past failures (Lecture 2.3b).